No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation	Predicts the most suitable crop using soil and climate parameters.	Prediction Module	Done	High
2	Machine Learning Models	Implemented Logistic Regression, Decision Tree, Random Forest, and KNN models.	ML Models	Done	High
3	Data Analysis	Performs dataset, univariate, bivariate, and multivariate analysis.	Analysis Module	Done	Medium
4	Data Preprocessing	Handles train-test split, seasonal analysis, and outlier detection.	Preprocessing	Done	High
5	Random Forest Prediction	Uses the best-performing model (99.32% accuracy) for prediction.	Prediction Engine	Done	High
6	Crop Information Display	Shows season, soil type, temperature, and water requirement.	Recommendation Page	Done	Medium
7	Responsive Web Interface	Provides an easy-to-use interface for users.	Frontend	Done	Medium
8	GitHub Version Control	Tracks project versions and collaboration.	GitHub	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Homepage, Prediction Page, Contact Page
2	Backend / Logic	4	Flask Routing, Prediction Logic, Crop Details, Validation
3	Database / Storage	2	CSV Dataset, Pickle Model Files
4	API / Integration	1	Flask Request Handling
5	Security / Authentication	1	Admin Login (Basic)